PROJECT BASED EXPERIENTIAL LEARNING REPORT
ON

"CardioSpike-FPGA: A Multiplier-Free Spiking Neural Network for On-Device ECG Arrhythmia Classification"

Under the Course
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SUBMITTED BY

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ABSTRACT

Cardiac arrhythmias represent one of the leading causes of sudden cardiac death globally, necessitating continuous monitoring solutions. While wearable ECG devices exist, they typically rely on cloud processing or power-hungry embedded processors, rendering true on-device, real-time edge classification a significant engineering challenge. This project, CardioSpike-FPGA, proposes a fully digital, multiplier-free Spiking Neural Network (SNN) deployed on a standard Field-Programmable Gate Array (FPGA) for real-time arrhythmia classification.

By leveraging the biological principles of Leaky Integrate-and-Fire (LIF) neurons and probabilistic rate coding, the architecture eliminates the need for hardware DSP multipliers by utilizing mathematically optimized bit-shift leak mechanisms. The network (topology 128 → 16 → 16 → 5) is trained offline using Surrogate Gradient Learning on the MIT-BIH Arrhythmia Database, mapping heartbeats to the 5 standard AAMI classes (N, S, V, F, Q). The resulting Q8.8 quantized fixed-point weights are hardcoded into FPGA Block RAM (BRAM), managed by a time-multiplexed Finite State Machine (FSM) that sequentially processes neurons to minimize Logic Look-Up Table (LUT) consumption. The proposed system aims to deliver highly competitive diagnostic accuracy while dramatically reducing power consumption compared to conventional Convolutional Neural Networks (CNNs), demonstrating a viable path forward for secure, battery-operated, extreme-edge medical wearables.

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1. INTRODUCTION

Cardiac arrhythmias are abnormal heart rhythms that represent one of the leading causes of sudden cardiac death and hospitalization globally. Early and continuous detection of arrhythmias can significantly improve patient outcomes, particularly for high-risk individuals who require round-the-clock cardiac monitoring outside of a clinical setting.

Traditional clinical ECG monitors are large, power-hungry devices confined to hospital environments. While wearable ECG devices exist, most rely on sending raw data to a cloud server or smartphone for processing, introducing latency and dependence on wireless connectivity. True on-device, real-time arrhythmia classification — where the computation happens directly on a low-power chip worn by the patient — remains a largely unsolved engineering challenge.

Spiking Neural Networks (SNNs) offer a biologically inspired computational paradigm ideally suited for this challenge. Unlike standard Artificial Neural Networks (ANNs), SNNs process information as discrete binary spike events. Because a spike is simply a 1 or a 0, multiplications are replaced by conditional additions — a property that maps directly and efficiently onto digital hardware fabric such as FPGAs.

This project, CardioSpike-FPGA, proposes the design and implementation of a fully digital, multiplier-free SNN-based ECG classifier on a standard commodity FPGA. The entire inference pipeline — from raw ECG input to arrhythmia class output — runs on the FPGA chip with zero dependence on an external processor or cloud.

2. PROBLEM STATEMENT

Current state-of-the-art approaches to on-device ECG classification face several critical limitations:

- **Excessive Power Consumption:** Recent high-accuracy neural network implementations consume upward of 330 milliwatts of power, draining wearable batteries in hours.
- **Reliance on SoC-Class Hardware:** Several FPGA implementations require expensive, power-heavy System-on-Chip (SoC) platforms with embedded ARM Cortex processors.
- **Hardware Multiplier Dependency:** Many "SNN-based" works rely on hybrid ANN-SNN architectures that still require power-hungry DSP slices on the FPGA.
- **Complex Pre-Processing Overhead:** Certain implementations require real-time computation of mathematical derivatives in hardware, adding latency and logic gates.

CardioSpike-FPGA addresses all four of these limitations simultaneously through a carefully co-designed software training pipeline and multiplier-free hardware architecture.

3. OBJECTIVES

- To design a 3-layer Leaky Integrate-and-Fire (LIF) Spiking Neural Network ($128 \rightarrow 16 \rightarrow 16 \rightarrow 5$), trained on the MIT-BIH Database to classify the five AAMI heartbeat categories.
- To implement a hardware-aware training pipeline that enforces a membrane leak rate of exactly $\beta = 0.9375$, guaranteeing that the LIF update reduces to a single arithmetic right-shift operation ($>> 4$), eliminating multipliers.
- To quantize all trained network weights to fixed-point format, enabling storage of all weights within a single Block RAM (BRAM) tile.
- To implement the complete SNN inference engine in VHDL, synthesizable on standard FPGAs, utilizing zero DSP slices.
- To validate the hardware implementation against the software model, confirming classification correctness and measuring FPGA resource utilization.

4. BRIEF LITERATURE REVIEW

The field of on-device neural ECG classification has grown significantly. The following works define the benchmark for this project:

- **Scrugli et al. (IEEE Sensors 2026):** Implements an SNN on a low-power Lattice FPGA using delta-modulation spike encoding. While highly power-efficient, it requires real-time derivative computation.

CardioSpike-FPGA eliminates this pre-processing overhead by using direct-amplitude LFSR rate coding.

- **Cao et al. (IEEE TVLSI 2026):** Achieves 99.5% accuracy using an Inception-ResNeXt CNN on FPGA. However, it relies heavily on DSP multipliers. CardioSpike-FPGA trades a small margin of accuracy to completely eliminate DSP usage.
- **Kachuee et al. (IEEE ICHI 2018):** Established the standardized 5-class AAMI mapping dataset extraction pipeline on MIT-BIH that serves as the benchmark dataset for all subsequent deep learning research.
- **Neftci et al. (IEEE SPM 2019):** Introduced Surrogate Gradient Learning, forming the mathematical foundation that allows the SNN in this project to be trained using backpropagation through time.

5. PROPOSED METHODOLOGY

Dataset & Pre-processing: The MIT-BIH Arrhythmia Database is utilized. R-peak locations are extracted, and a 128-sample window (355 ms) is centered on each peak. Windows are min-max normalized to a [0,1] range and mapped to 5 AAMI classes. Data augmentation (noise, shift, scaling) corrects severe class imbalances.

Network Architecture & Training: A 128 → 16 → 16 → 5 dense architecture is trained in Python using snnTorch (Surrogate Gradients, BPTT over 16 timesteps). The network utilizes zero-reset LIF neurons with a hardware-friendly leak rate.

Hardware Design (VHDL): The inference engine is designed as a time-multiplexed Finite State Machine (FSM). An 8-bit LFSR generates random thresholds for rate-coding the normalized inputs into spikes. A sequential multiplier-free MAC unit (conditional addition + bit-shift leak) processes the hidden layers, and an Argmax combinational circuit determines the final diagnosis.

6. APPLICATIONS

- **Wearable Cardiac Patches:** The ultra-low power budget enables weeks of continuous ambulatory ECG monitoring on a coin-cell battery.
- **Remote Patient Monitoring:** Eliminates cloud data transmission, making cardiac monitoring accessible in low-resource or rural settings.
- **Secure Defense & Aerospace:** Hardware-isolated inference provides extreme reliability and security against cyber threats for high-value personnel.

7. REFERENCES

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